EXPERIMENT - 12

Security, Governance and Administration

Library Book Issue Management

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Subject	Low-Code Application Development using Zoho Creator
Application	Library Book Issue Management
Platform	Zoho Creator
Record Date	16 October 2026

Aim

To implement and verify security and governance practices for the library application through role-based permissions, controlled administrative access, audit records, safe integration handling and backup management.

Experiment overview

Review and verification of least-privilege profiles, audit history, integration access and recoverable project exports.

Department of Computer Science and Business Systems

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1. Security Requirements

Members should submit requests and see permitted own records. Staff should process the queue without unrestricted administration. The owner should manage application design, integrations and release decisions.

2. Permission Matrix

Capability	Library Admin	Library Staff	Library Member
Submit request	Yes	Yes	Yes
View requests	All permitted app data	Processing reports	Permitted own records
Edit request / status	Yes	Yes	No
Delete / export	Administrative profile	No	No
Dashboard	Yes	Yes	No
API access	Administrative control	Disabled	Disabled
Design / release	Owner / developer authority	No	No

Roles describe hierarchy; permission profiles control component actions; data sharing controls record visibility. A role name by itself does not enforce all three layers.

3. Verification Procedure

1. Opened Settings > User Permissions and inspected the three named profiles.

2. Checked Library Member: form access enabled, edit/delete disabled and dashboard access disabled.

3. Confirmed that the Member API-access option is Disabled.

4. Reviewed earlier field-permission and role-simulation evidence.

5. Opened Operations > Audit Trail, selected the library app and Book_Issue_Request_Form, and inspected actual events.

6. Verified the downloaded schema/data and official Production backup files.

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4. Governance Controls

Control	Project implementation / observation
Least privilege	Separate Admin, Staff and Member profiles. Members cannot process requests.
Field restrictions	Request_Status is read-only for Members. Staff_Notes and Confirmation_Sent are hidden from Members in the earlier configured profile.
Data isolation	Earlier role simulation checked own-record visibility. A separate authenticated account session has not been completed.
Audit trail	Actual create/edit entries, timestamps, actors and record IDs were inspected on 16 October. Public submissions appear with a Public actor.
Integration secrets	Zoho Mail uses a named connection. Do not put tokens or passwords in screenshots or exported documentation.
Notification state	Confirmation_Sent records the application outcome; it is not proof that a recipient read or received the message.
Release governance	Development is separated from Stage and Production. New changes require package review.
Recovery	Official Production backup, schema exports and CSV files retained locally.

5. Administrative Routine

Review active users and their assigned profile at the end of the lab. Remove unnecessary sharing only with authorization. Check failed workflows and unexpected audit entries, retain a dated backup, and document approved changes.

No password, MFA, domain restriction, organization-wide policy or paid security setting was changed for this experiment. The existing public form is a submission surface, so anonymous submissions must not be treated as verified member identity.

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Implementation Evidence

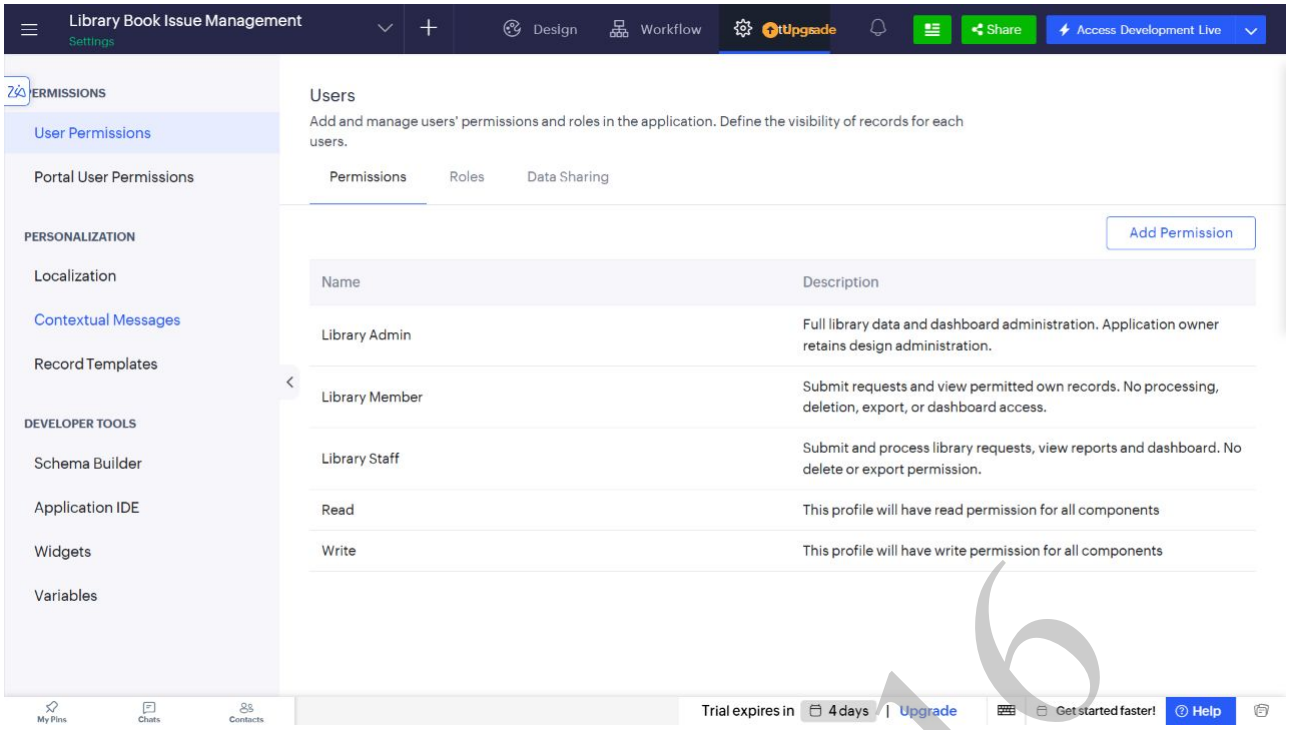


Figure 12.1. The existing Library Admin, Library Staff and Library Member permission profiles, inspected 16 October.

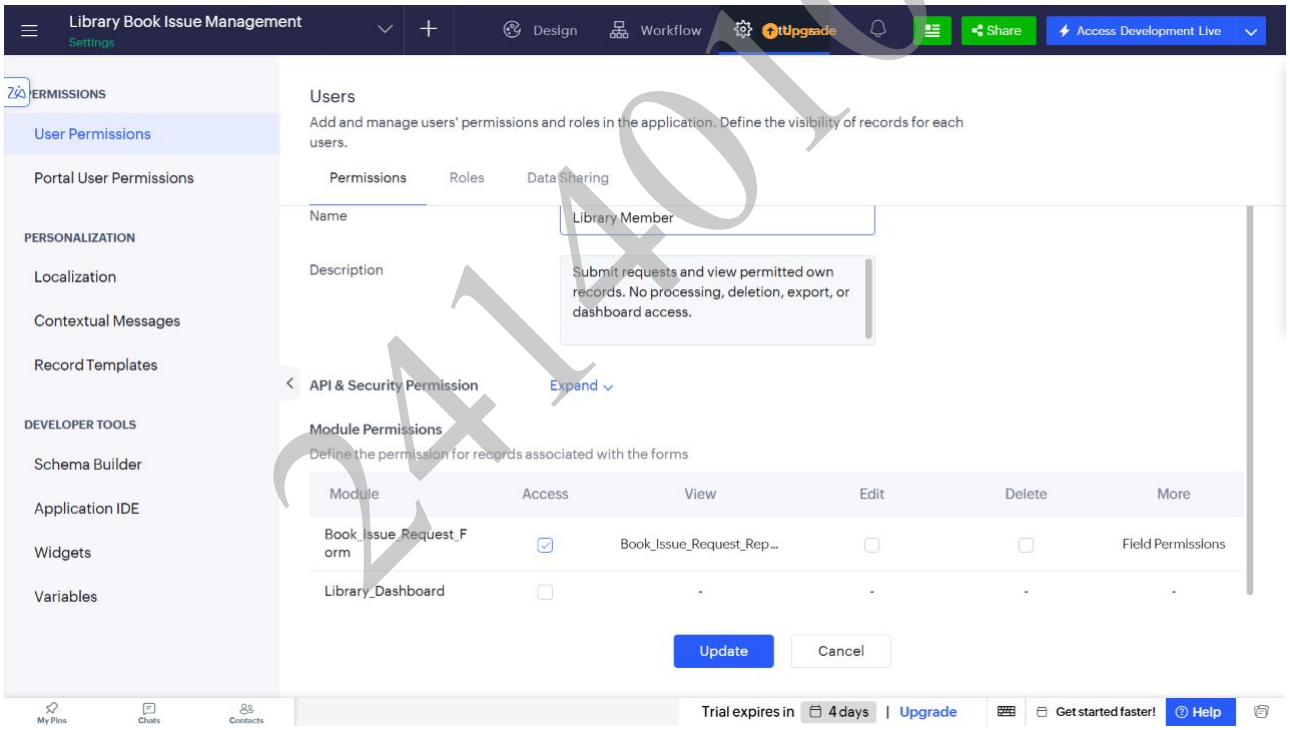


Figure 12.2. Member profile inspection: no edit/delete/dashboard permission and API access disabled.

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Implementation Evidence

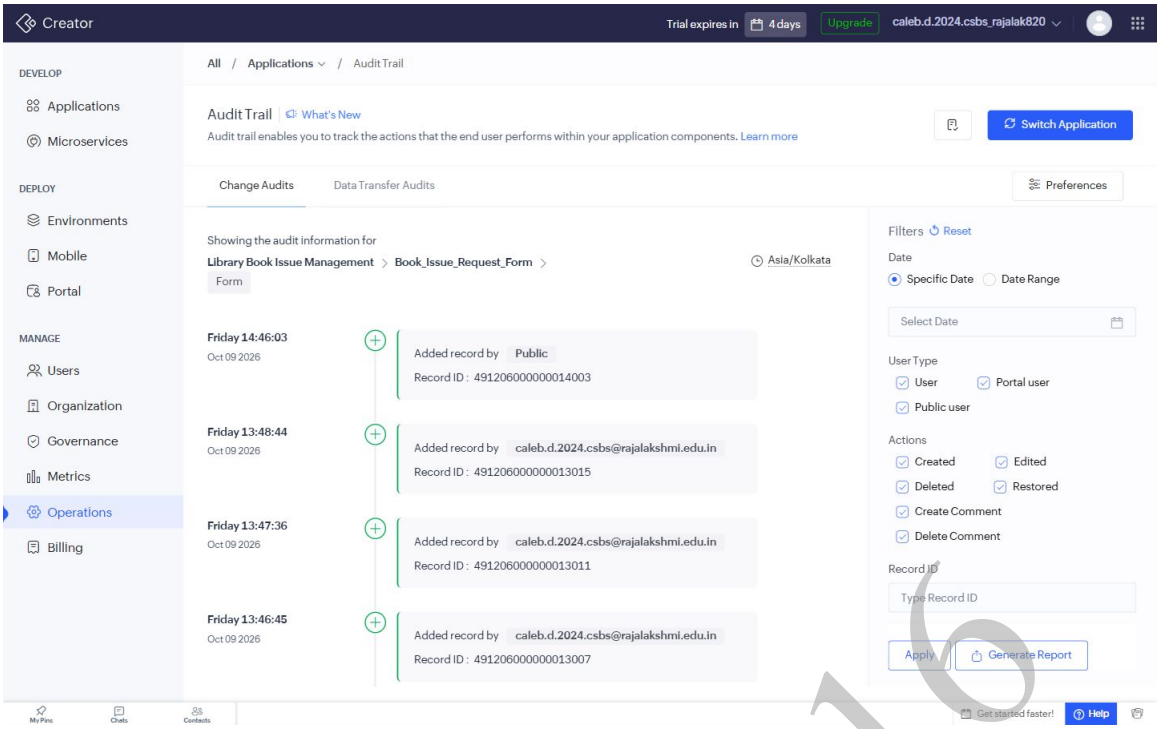


Figure 12.3. Actual library request audit trail. Events include actor, date/time, action and record ID.

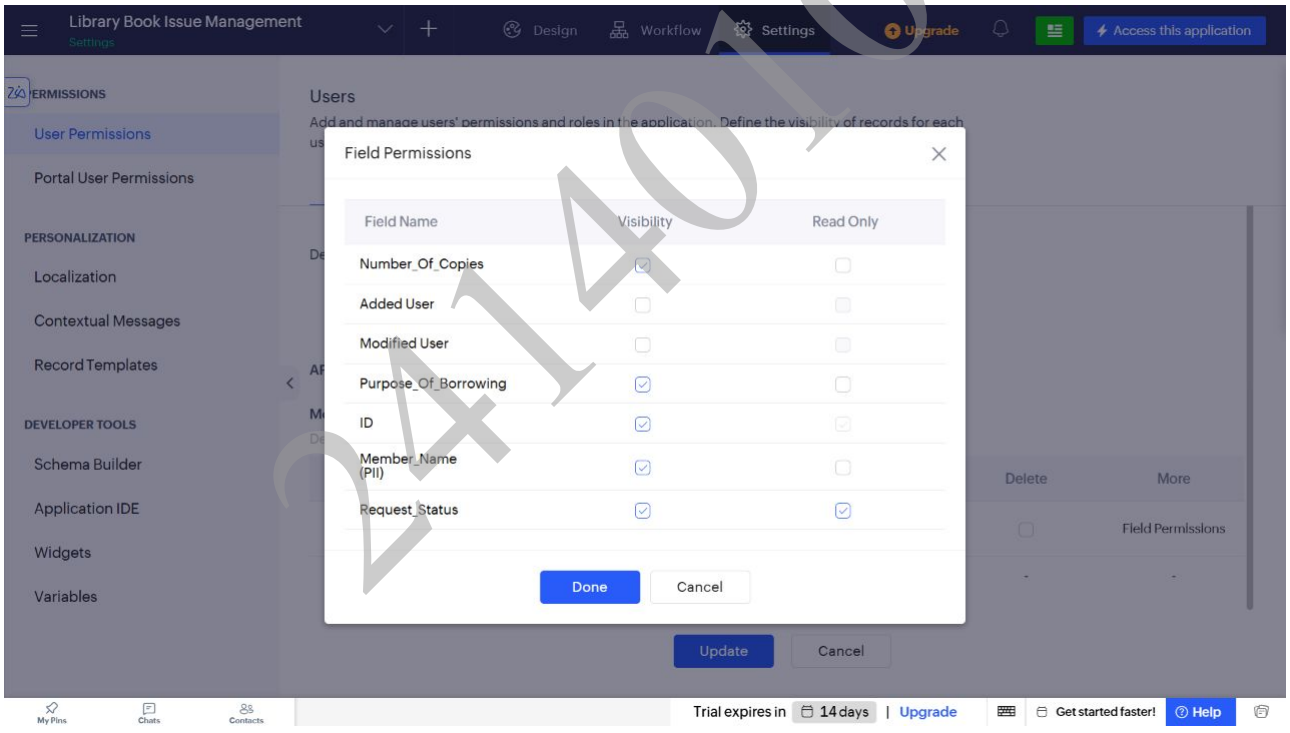


Figure 12.4. Preserved field-level permission evidence from the earlier RBAC configuration; not a separate-account login test.

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6. Result

The project has a defined least-privilege permission model, inspectable audit events and a downloaded backup set. Existing controls were reviewed without broadening access. Governance procedures connect permissions, notifications and releases to named responsibilities.

7. Verification Limits

Item	Status
Named profiles and visible Member restrictions	Verified in the current UI
Audit history	Real events inspected
Role simulation	Earlier evidence retained
Real separate-account login	Not verified in this run
Backup restoration	Archive retained; isolated restore not performed
Formal compliance certification	Not assessed; this is a student lab

References

1. Understanding user permissions

2. View and manage audit trail

Official sources checked on 16 October 2026. App observations and test results are identified separately.

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